

detectors can detect only syntactically similar code blocks, and not semantically similar ones (which we discussed a bit earlier). Even with this disadvantage, it is tempting to use clone detectors because most of them are:

- 1) Programming-language-agnostic, or handle multiple programming languages;
- 2) Quite fast—they often process millions of lines of code in a few hours;
- 3) Fairly precise in finding code clones, with only a small percentage of ‘false alarms’;
- 4) Command-line tools and can be integrated as part of regular builds (for regular monitoring);
- 5) Free or quite affordable (unlike costly static and dynamic code analysers for detecting bugs).

There are specific kinds of code clone detectors that are extraordinarily costly! Can you guess why?

Given a tight deadline, programmers working in software companies sometimes copy code segments they get through search engines, and use these in the code they develop. Of course, this is a dangerous practice; the consequences of copying from open source software to an enterprise software code base can be serious. So,

software companies want to make sure that they are safe, and their code base is free of any code segments from open source—for which they purchase specialised clone detector tools, which can be quite expensive!

How do these tools work? They maintain a database of open source software, and apply sophisticated clone detection techniques to check if there are any copies of open source code segments in the given code base. Real world enterprise software code bases often exceed a million lines of code. Because the database of open source software is so vast, and the tool has to check for small code segments from the large enterprise code base, it takes many days (or even weeks) to complete running such tools! Because of their specialised nature, these clone detector tools are extremely costly and there is a micro-industry of companies selling these tools!

So, do you now agree that using *Ctrl+C* and *Ctrl+V* to meet that all-too-important deadline is dangerous? 

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